THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 9 — SCIENCE FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–13) | RBSE Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Draw neat, labelled diagrams wherever required.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. Climate change and designing sustainable technology need ideas from: 1

(a) several branches of science together **(b)** only biology
(c) no science **(d)** only physics

2. The three main aims when approaching a numerical science problem are to understand the situation, identify the quantities that matter and then: 1

(a) skip units
(b) use relationships and check with an estimate
(c) guess the final value
(d) memorise answers

3. The size of the particles in a suspension is: 1

(a) smaller than 1 nm **(b)** equal to atoms
(c) too small to filter **(d)** large enough to be seen and to settle

4. The Tyndall effect would be seen MOST clearly in: 1

(a) pure alcohol **(b)** sugar solution
(c) clear water **(d)** a beam of light passing through fog

5. The correct name of the covalent compound PCl_3 is: 1

(a) phosphorus trichloride **(b)** phosphorus chloride
(c) monophosphorus trichloride **(d)** phosphoric chloride

6. A formula unit mass is used for: 1

(a) gases only **(b)** elements only
(c) ionic compounds **(d)** covalent compounds

7. The maximum number of electrons that the shell with n = 4 can hold is: 1

(a) 64 **(b)** 34 **(c)** 16 **(d)** 32

8. A cyclist covers 900 m in 50 s along a straight road without turning back. The average speed is: 1

(a) 9 m s^{-1} **(b)** 18 m s^{-1} **(c)** 36 m s^{-1} **(d)** 0.06 m s^{-1}

9. The slope of a velocity–time graph gives: 1

(a) speed **(b)** velocity **(c)** displacement **(d)** acceleration

10. The momentum of a 2 kg ball moving at 5 m s^{-1} is: 1

(a) 25 kg m s^{-1} **(b)** 2.5 kg m s^{-1} **(c)** 7 kg m s^{-1} **(d)** 10 kg m s^{-1}

11. A net force of 100 N acts on a body of mass 25 kg. Its acceleration is: 1

(a) 4 m s^{-2} **(b)** 2500 m s^{-2} **(c)** 0.25 m s^{-2} **(d)** 8 m s^{-2}

12. A constant force of 120 N displaces a body by 4 m in the direction of the force. The work done is: 1

(a) 960 J **(b)** 124 J **(c)** 30 J **(d)** 480 J

13. At a point in a medium, 120 density oscillations occur in 6 s. The frequency of the sound wave is: 1

(a) 40 Hz **(b)** 20 Hz **(c)** 0.05 Hz **(d)** 126 Hz

14. The 'suicide bags' of the cell, containing digestive enzymes, are: 1

(a) vacuoles **(b)** lysosomes **(c)** ribosomes **(d)** plastids

15. The cell wall in plants is mainly made of: 1

(a) cellulose **(b)** lipid **(c)** protein **(d)** chitin only

16. Bones are connected to bones by: 1

(a) blood vessels **(b)** ligaments **(c)** muscles **(d)** tendons

17. Starfish and sea urchins belong to: 1

(a) Echinodermata **(b)** Mollusca **(c)** Porifera **(d)** Cnidaria

18. The male gametes in humans are: 1

(a) spores **(b)** sperms **(c)** pollen grains **(d)** ova

19. Which method is used for growing roses and sugarcane from stem pieces? 1

(a) budding **(b)** cutting **(c)** spore formation **(d)** pollination

20. If the insolation is 1 kW m^{-2}, the solar energy received by 10 m^2 in 4 h is: 1

(a) 14 kWh **(b)** 80 kWh **(c)** 40 kWh **(d)** 43 kWh

SECTION — B (Very Short Answer Type) — 8 × 2 = 16

Answer in about 10–15 words each.

21. Give one example where an approximate answer is enough and one where an exact value is needed. 2

22. How does sodium chloride form? Write the ions. 2

23. Why is circular motion at constant speed accelerated? 2

24. Why is it difficult to walk on a wet polished floor? 2

25. State the function of lysosomes. 2

26. What are the functions of epithelial tissue? 2

27. Differentiate between external and internal fertilisation. 2

28. How does the nitrogen in the atmosphere become useful to plants? 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 30–40 words each.

29. Why do carbonated drinks lose their fizz when kept warm? Relate to solubility. 3

30. Name the three isotopes of hydrogen and give the number of protons and neutrons in each. 3

31. Classify the levers: scissors, wheelbarrow and tweezers. 3

32. Explain why we hear echoes in a large hall but not in a small room. 3

33. Why are cardiac muscles specialised? How do they differ from skeletal muscles? 3

34. Distinguish platyhelminthes from nematodes. 3

35. Explain the formation of magnesium chloride. 3

OR

Why does NaCl conduct electricity in solution but sugar does not? 3

36. If photosynthesis stopped on Earth, what would happen? 3

OR

Why is the rise in atmospheric CO_2 a concern? Mention two effects. 3

SECTION — D (Long Answer Type) 4 × 5 = 20

Answer in about 100–150 words each.

37. (i) State the Law of Constant Proportions and name the scientist. 5
(ii) Water from a river and from the sea each give H : O = 1 : 8. Explain.
(iii) 36 g of water is decomposed; find the masses of hydrogen and oxygen.
(iv) Does the law hold for mixtures?

OR

(i) What is a covalent bond? 5
(ii) Show the formation of H_2, Cl_2 and HCl by sharing of electrons.
(iii) How many shared pairs are there in O_2 and N_2?

38. **Case study:** A student pushes a 25 kg block; maximum friction is 50 N. 5
(i) What happens with a 50 N push?
(ii) What is the net force with 55 N?
(iii) Find the acceleration.
(iv) Find its displacement in 2 s.

OR

(i) Describe Activity 6.3 to show that acceleration depends on force. 5
(ii) How does mass affect acceleration?
(iii) Write the law in the mathematical form.

39. (i) What is the role of the nucleus? 5
(ii) Describe chromatin, chromosomes and genes.
(iii) Why can mature RBCs not divide?

OR

(i) State the cell theory. 5
(ii) Who contributed to it?
(iii) Why is the cell called the basic unit of life?

40. (i) Describe pollination and its types. 5
(ii) Explain the agents of pollination with examples.
(iii) What happens after pollination?

OR

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Case study: Bread left in a moist, warm place grows fuzzy black patches. 5

(i) Name the organism.

(ii) How did it reach the bread?

(iii) How does it reproduce?

(iv) State two uses of fungi.

END OF QUESTION PAPER **Total: 80 Marks**

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